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Resolving Complex Subsurface Integrity Issue Through Monobore Completion: A Success Story of Idle Injector Reactivation Pilot to Support Development in Merdeka Field

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Abstract

Merdeka field has been producing with waterflood for more than 50 years. Aging infrastructures and increasing well integrity challenges have led to a growing number of idle injector wells, mostly due to casing leaks caused by corrosion.

This paper presents a pilot study in which a previously idle injector was successfully reactivated using a monobore remedial completion. The reactivation was part of a comprehensive subsurface reservoir management plan aimed at restoring injectivity and improving the waterflood performance on a mature waterflood field with active development activities.

The study outlines the decision-making process, subsurface review, candidate selection, well intervention design and post-reactivation impact evaluation. As a result, the pilot well was successfully reactivated, achieving a stable injection rate of 23,000 BWIPD at 500 psi, which led to a 15% increase in production from the surrounding producers and 7% improvement of VRR in the respective field compartment. The results offer a framework for improving waterflood performance in mature waterflood fields through an innovative method of idle injector reactivation.

Introduction

Waterflooding has been one of the main methods of secondary oil recovery for decades, contributing to improved sweep efficiency and delayed production decline. However, aging conventional injector wells often introduce mechanical complexity and increasing maintenance costs.

Monobore wells provide a solution that increases injection efficiency while reducing operational risks. (Etuhoko et al, 2004). The cost reduction achieved through the simplified well design enhances the method's economic feasibility by lowering both capital and operational cost. The reduced requirement for conventional well interventions creates a more efficient and reliable field development strategy. The potential to eliminate several conventional workovers using monobore completions helps reduce operational risks, such rig delays, cost overruns and safety incidents (Macfarlane et al, 1998)

Mature fields often struggle with depleting reservoir pressure and uneven sweep efficiency. A comprehensive injection strategy is fundamental to sustaining oil production performance, particularly in heterogeneous or pressure depleted reservoir within the field. Mechanical wellbore issues, zonal communication or casing leaks often make aging injector optimization become challenging. This paper presents a pilot study where an old injector with complex integrity issues was restored using a monobore remedial completion. The injector reactivation is aligned with a subsurface strategy to optimize injections and reduce the requirement for drilling new injectors. The project integrates subsurface mapping, zonal prioritization, remedial well design, post-job result monitoring and economic evaluation.

Background

Merdeka field is a large field which extends across a large anticlinal structure with an average reservoir depth of 2200 ft. The field was segmented into multiple compartments separated by faults. Most of the reservoir have relatively good reservoir properties such as continuity, thickness, porosity and permeability. The field has been under waterflood since 1971 and at present has reached more than 50% oil recovery. All the producer wells are operating by using ESP (Electric Submersible Pump) as the artificial lift system. Currently the field is at mature waterflood stage with high water cut production.

One of the main focuses of the waterflood operation is to replace the volume of fluids that are extracted from the reservoir with continuous injection where Voidage Replacement Ratio (VRR) being monitored as the key performance metric. A VRR of less than 1.0 implies that the reservoir is being depleted or under injected. On the other hand, a VRR much greater than one may indicate over-injection, leading to early water breakthrough or fracturing of the formation. Effective monitoring of VRR helps optimize production and maintain reservoir integrity (Ranjith et al, 2017).

Maintaining VRR in large-scale waterflood operations such as Merdeka field is a complex task. In reservoirs with permeability variations and compartmentalization, achieving uniform sweep efficiency becomes very challenging. This often results in uneven injection distribution, where certain zones experience over-injection and premature water breakthrough, while other areas of the field remain under-injected and ineffectively swept. The large number of wells in large fields, combined with varying producer-to-injector ratios across different areas makes managing injection volumes and timing difficult. This often results in deviation from the desired VRR and leading to poor waterflood performance and low oil recovery.

Merdeka field currently operates with more than 1000 active producers and injectors. It operates with zero external water discharge policy, which means all the produced water must be injected back into the reservoir. The field is segmented into multiple compartments, making it highly challenging to maintain consistent VRR across all compartments while managing dynamic production and injection operational condition in the field. As illustrated in Table 1, VRR varies significantly between field compartments, where some compartments indicate over-injection condition and others demonstrate under-injection tendency.

Table 1—VRR Performance in Field Compartments of Merdeka Field

Field Compartment	VRR	Field Compartment	VRR
COM-1	>1	COM-6	<1
COM-2	>1	COM-7	<1
COM-3	>1	COM-8	<1
COM-4	>1	COM-9	>1
COM-5	<1	COM-10	>1
		COM-11	<1

Another issue that is common in mature waterflood fields, which has been encountered in Merdeka field, is the aging infrastructure. Increasing integrity challenges tend to increase overtime and leads to a growing number of idle injector wells mostly due to casing leaks.

As fluid production increased with significant number of new wells and fast field development activities in Merdeka field in recent years, reactivating these idle injector wells became increasingly crucial. It is important to improve fluid balance in all the reservoir across the field, especially in under injected areas. With the zero-water discharge policy, injector reactivation is also important from surface facility perspective to increase injection point in order to avoid excessive pressure on injection line and mitigate pipeline bottlenecks.

Injector reactivation campaign has become one of the focus activities in Merdeka field in recent years. In many reactivation cases, conventional methods like casing sleeve and squeeze cementing have succeeded in resolving casing leak issues found on majority of idle injectors. However, this method failed for wells with complex subsurface integrity issues such as very deep leak points or multiple leak points. This study was conducted to implement alternative completions to solve the issue using integrated G&G data, well integrity assessments, and workover history evaluation.

Method

Screening Process

A screening process has been established to identify the most suitable injector wells for reactivation campaign in the Merdeka Field. The evaluation considers many factors from subsurface data, injector well history, producer data and surface facility constraint.

1. VRR (Voidage Replacement Ratio) Performance

   An effective waterflood relies on carefully controlled and balanced water injection to efficiently drive oil toward the production wells (Craft et al., 1991). Field compartment with VRR significantly above 1 is usually an indication for over-injection, leading to over-pressure zones and water cycling between wells. Injector reactivation should be focused on areas with relatively lower VRR to maintain reservoir pressure on the area while redistributing injections across the field properly.
2. Producer to Injector Ratio

   Injectors located in fast developing areas or areas with dense well population often produce suboptimal injection performance. Reactivation efforts should prioritize injectors located on these areas, particularly where producer to injector ratio is significantly high, to help restore fluid balance in the field.
3. Connectivity to Nearby Producers

   Accurate geological interpretation is critical to waterflood operations, as factors such as reservoir heterogeneity, faulting, and stratigraphic barriers directly affect fluid flow and sweep efficiency. Strong communication between injectors and nearby producers is essential to ensure effective waterflood management and maximize recovery (Wardhana et al., 2019). Injector reactivation should only proceed where established pressure support contributes directly to nearby production performance. The evaluation process was conducted by evaluating well cross sections between the injector with its surrounding producer and production performance from historical data of the producers in the past when the injector was still active
4. Well Location

   Injectors located in updip location or higher areas relative to surrounding producers often caused early water breakthroughs. Lower structural injectors typically offer greater potential for volumetric sweep in layered reservoir system (Bui et al, 2010). Therefore, reactivation efforts should prioritize injectors that are located on downdip or lower area relative to its surrounding producers

5. Wellbore Integrity

   Wells with a history of casing deformation or complex fishing history must undergo comprehensive evaluation before being reactivated. Reactivating wells with major well integrity issues or repetitive leaking history without proper mitigation plan can lead to uncontrolled fluid losses or inability to maintain target injection rates. Risk mitigation includes evaluating cement bond logs and performing pressure integrity tests to confirm mechanical reliability.
6. Fluid Level Observations

   Frequent shallow static or working fluid levels in surrounding producers often indicate inefficient water injection. Wells with shallow fluid levels, especially in areas with minimum withdrawal, may not benefit from further additional injections as it might cause increasing water cycling tendency with minimum improved oil recovery. In some extreme cases, excessive injections into the reservoir can result in issues such as over pressured zones and uncontrolled fluid flowing to the surface. Injector reactivation efforts should prioritize areas where shallow fluid levels are uncommon, or areas with active withdrawal and ongoing field development activities.
7. Surface Facility Constraint

   High-pressure injection lines often indicate surface facility nearing its capacity limits. In fields with aging facilities, careful assessment of surface injection capacity is as important as subsurface evaluation. Injector reactivation should be focused on redistributing the injection to areas where additional pressure in injection lines is beneficial for the field injection system, both from reservoir and surface facility operational perspective. Injection line pressure data and the surface injection system model were also utilized to identify bottlenecks or segments of the injection network experiencing excessive pressure. The injector reactivation campaign should be strategically designed to address and resolve these constraints. Another necessary risk mitigation measure is to conduct early surveys to assess the condition and completeness of the injector and its supporting surface facilities, especially after being idle for many years.

Well Classification and Planning

Following the screening process, many idle injectors satisfying reactivation criteria have been identified. Subsequent efforts will focus on developing optimized reactivation procedures for efficient execution. The initial phase involves a comprehensive assessment of historical workovers and operational reports to determine the root causes of the injector being shut-in in the past. Common causes for idle injectors in Merdeka field include:

1. Surface facility issues e.g leaking injection line or valves
2. Subsurface integrity issue in adjacent producers, e.g fluid flowing in the surface
3. Wellhead or injection packer leak
4. Shallow casing leak
5. Deep casing leaks or multiple leak points

Following the candidate classification process, an injector reactivation campaign was initiated by engaging working teams including drilling and well intervention teams, facilities and operation teams. The execution procedure varies according to each injector issue and its operational complexity. Less complex cases may only involve remedial effort from facility and operation team, such as repairing broken injection lines or valves. For shallow casing leaks, rig less remediation techniques such as casing sleeve installation or top job are employed. For well with deeper casing leak or well head leak cases, typically require rig intervention jobs.

In specific cases involving very deep casing leaks or multiple leak points, a comprehensive analysis and coordination process was conducted to evaluate feasible remediation alternatives. Historically attempted interventions under such complex conditions have produced unsuccessful results in Merdeka field.

Alignment with Field Development Strategy

Merdeka Field has experienced significant shifts in its development strategy. Following many years of no drilling new well activities, field development activities have resumed at an average total of 50 new wells annually in recent years.

This strategic realignment has been driven by updated G&G data and good oil prices, leading to a more aggressive approach and strategically focused development in the field

The result of the strategy is quite evident, as indicated by the flattening field decline rate in recent years. However, at the same time, the rapid increase of fluid production in certain areas has caused imbalanced waterflood operation. Some areas tend to have under injections resulting from the fast increase of new producer wells that are not balanced with the increase of injection points in the area. This caused problems such as the decreasing fluid level and declining productivity. Due to the limited active injectors combined with zero water discharge constraint, the inability to redistribute injections properly across the field have caused over injections in the nearby area with relatively less reservoir withdrawal or development activities. This led to problems such as over pressured reservoir and leaking producers with well integrity issues. The unresolved high injection pressure issue has led to the shut-in of several producers to reduce excessive fluid incoming to the surface gathering station, which is limited by fluid handling and injection capacity constraints.

Recent surveillance data identified under-injected areas in field compartment COM-5. The average Producer to Injector ratio in the compartment is 4, which is relatively high compared to other field compartments in Merdeka fields. This condition correlates with relatively lower VRR compared to other field compartments as previously shown in Table 1. VRR historical performance as shown in Figure 1 and declining reservoir pressure shown in Figure 2 also demonstrate that additional injection support is currently required in compartment COM-5

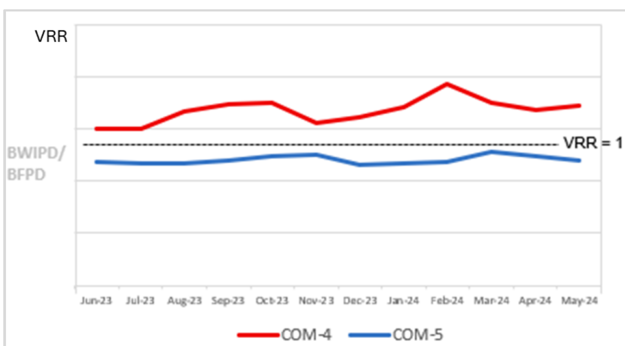

Figure 1—VRR Performance Comparison

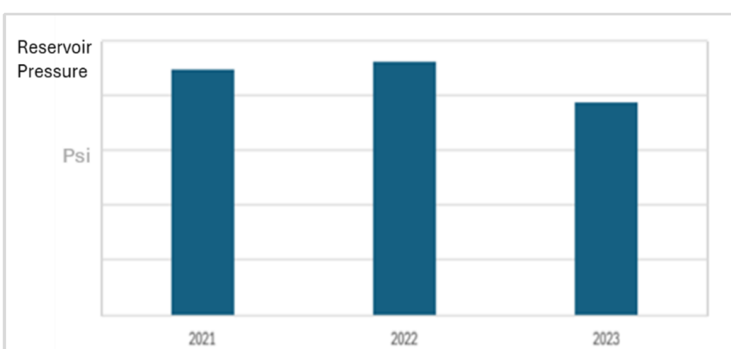

Figure 2—Annual Average Reservoir Pressure Trend in Field Compartment COM-5

One of the idle injectors in the compartment COM-5 is INJ-1. The injector has been active since 1999, before it was being shut in at 2018 due to casing leakage. Surrounding active producers mostly produced from respective P, Q and R reservoir flow unit and generally located relatively more updip or higher area compared to the injector. Producers with shallow fluid levels are rarely encountered in the nearby area. Current VRR data shows that the corresponding sand P, Q and R flow units need more injection support after significant period of active field development activities within the area. Recent zonal pressure map shown that reservoir pressure is depleting in areas nearby the injector as shown in Figure 3.

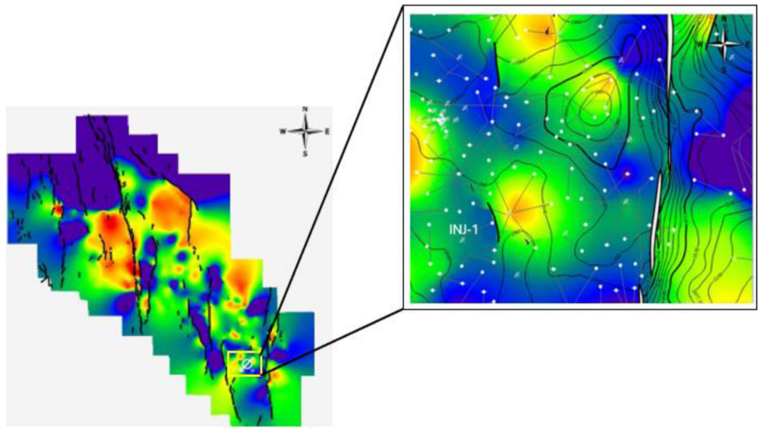

Figure 3—Reservoir Pressure Map Nearby INJ-1

Well injection data in the past and nearby producer's production data history demonstrate that the injector exhibits strong reservoir connectivity with multiple active producers in the surrounding area. Reservoir KH (Permeability x Thickness) map also indicates that generally areas nearby INJ-1 can be categorized as high injectivity reservoir, as shown in Figure 4

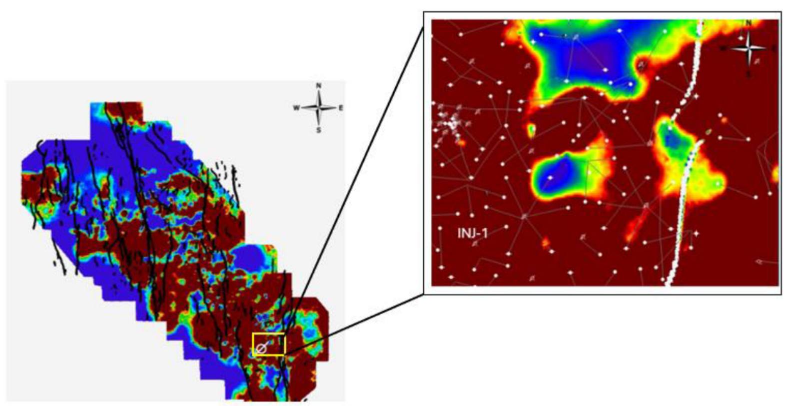

Figure 4—Reservoir KH (Permeability x Thickness) Map Nearby INJ-1

Well correlation analysis is shown in Figure 5. In this area, the P flow unit consists of blocky sand, while Q and R flow units exhibit variations in sand thickness. However, all flow units have good continuity with insignificant facies changes since they were deposited in a relatively uniform tidally influenced fluvial-deltaic environment. Reactivation presents an opportunity to improve production support, for example to producer wells PROD-3 and PROD 4 shown in Figure 5, by adding injection support in P, Q and R flow unit.

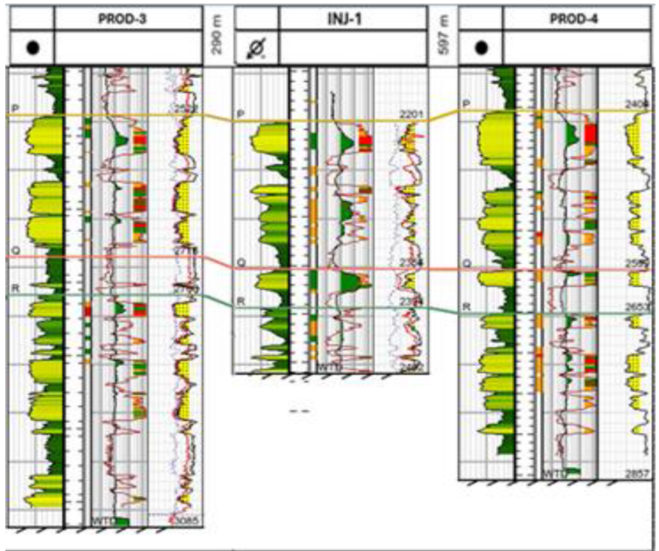

Figure 5—Well Correlation in Surrounding Pilot Monobore Injector INJ-1

Well Intervention Design and Execution

Production casing leaks is a well-integrity problem which can be fixed by circulating cement through the leak points until the cement returns to the surface, a technique known as the bandage method (Kurniawan, 2023). The job sequences for filling cement in the double casing annulus are as follows: (1) Barrier Installation, (2) Communication Test, (3) Cement circulation, (4) Shut in the annulus, and (5) Waiting on Cement.

When the leak point is below the surface casing shoe, improving well integrity is more challenging, particularly in Merdeka field. Multiple attempts to circulate cement out to the surface were unsuccessful. Cement circulation is most likely to be attained at leaking point above the surface casing shoe.

INJ-1 had been idle since 2018 due to casing leak at 1,392 ft, which is below surface casing shoe. Since the injector well has been in operation for 19 years, the surface casing was observed to be in a deteriorating condition as well. This injector reactivation candidate was considered to have complex well integrity issues which could not be resolved with conventional squeeze method.

After in depth review and risk assessments, monobore remedial completion was selected as the alternative to recover the well integrity. The monobore completion is a type of wellbore construction with a uniform small casing diameter and cemented behind the casing. In some cases, the well used a smaller casing since the initial well completion, because there are no plans to run in hole a large piece of equipment into the wellbore, such as mechanical packers or conventional artificial lift equipment. In other cases, the use of monobore completion for well reactivation has been carried out by placing cement in the annulus, effectively converting the tubing to become the new tubular or production casing.

The trial of monobore recompletion for this pilot study in Merdeka field was executed in four stages as follows:

1. Isolate perforation interval with cement plug

   The cement barrier shall be in place during temporary suspension prior to changing wellhead job. A verification test is conducted for well control assurance.
2. Change Out the Wellhead

   The wellhead is designed to enable injection fluid flow directly into the injection tubing by threaded connection inside the tubing hanger in order to avoid potential erosion of the rubber seal which is typically found in conventional injector wellhead.
3. Drill out cement, Run In Hole new tubular tubing 4-1/2", and Cementing 4-1/2" × 7" annulus

The new 4.5" tubing is set inside the old 7" production casing then cemented to establish a new well barrier in the annulus.

4. Perforation Job in target interval on P, Q and R sand

   The perforation job was carried out with 3-1/8" deep penetration gun with a density of 6 shots per foot (SPF), resulting in penetration of approximately 2 ft into the formation.

The well schematic before and after the job is depicted on Figure 6

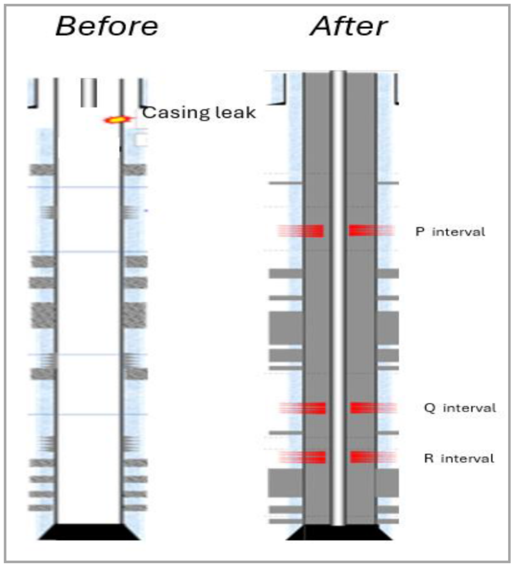

Figure 6—Well Schematic of Pilot Monobore Injector INJ-1

Result

Injection performance and Production Response

The well intervention was executed successfully. Regular weekly observations were conducted as an integral part of the pilot. Flow rate and injection pressure data show consistent flow at an average injection rate of 23,000 BWIPD, maintained at a stable wellhead pressure of 500 psi, as depicted on Figure 7.

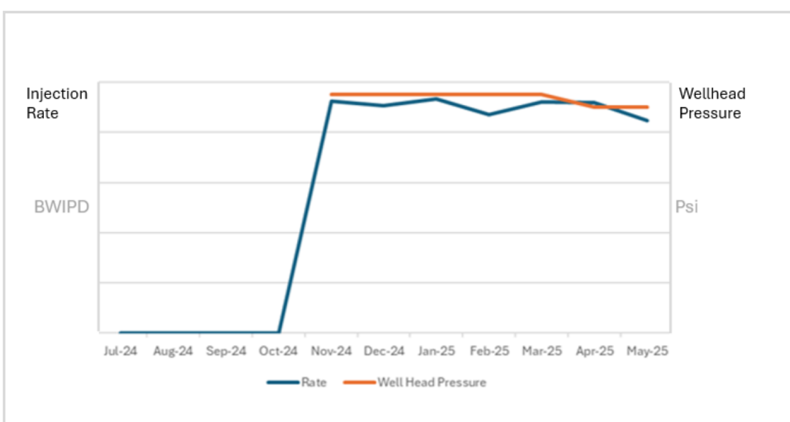

Figure 7—Injection Performance Monitoring of Pilot Monobore Injector INJ-1

From the field production perspective, an increasing trend of fluid level was observed on surrounding producers after the injector reactivation in Oct 2024. An example fluid level data monitoring from producer PROD-4 is shown in Figure 8. The data shows the increasing level of fluid column above the pump intake.

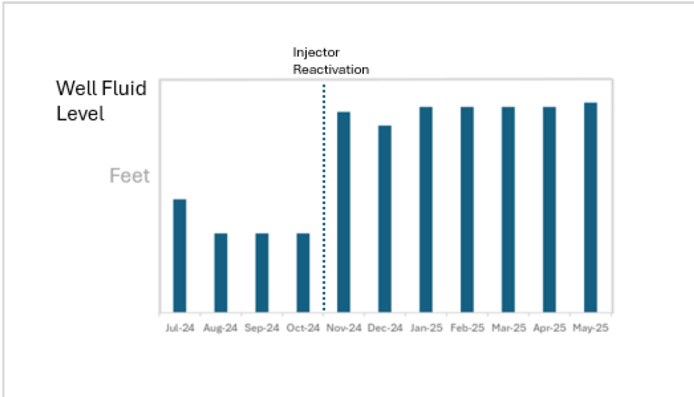

Figure 8—Well Fluid Level on Producer PROD-4 after INJ-1 reactivation

This trend suggests improved reservoir pressure support in the surrounding area, which has led to an observed increase of VRR in field compartment COM-5 by 7% and oil production increase from nearby producers by 15% as shown in Fig.9 and Fig 10

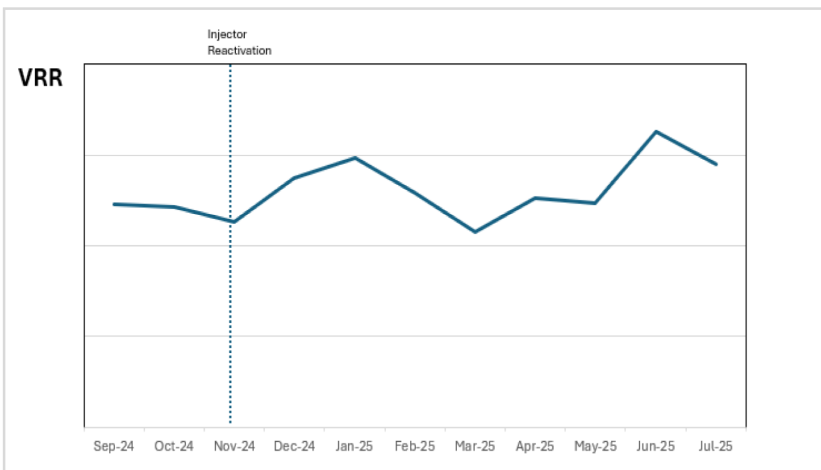

Figure 9—VRR Performance on COM-5 after INJ-1 reactivation

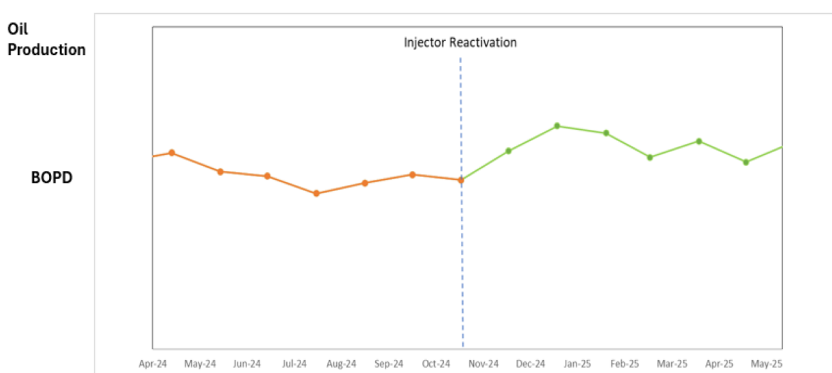

Figure 10—Surrounding Producer Production after INJ-1 reactivation

Additionally, improvements have also been observed in surface production system. Previously, several producer wells were shut in due to injection capacity constraints at the respective gathering station. Following the injector reactivation, the number of shut-in wells has shown a declining trend, indicating a positive impact on the performance of the surface injection and production system. Several producers have been brought back online, contributing to the improvement in field production performance.

Economic Review

Monobore recompletion is considered significantly more cost-effective compared to drilling new injector wells or converting producers into injectors. The actual cost for the pilot monobore recompletion is only 8% of the cost of drilling new injector well or 31% of the cost of converting producer to injector in Merdeka field. Comparison of the cost of monobore remedial completion with average cost of other injector alternatives in Merdeka field is shown in Figure 11.

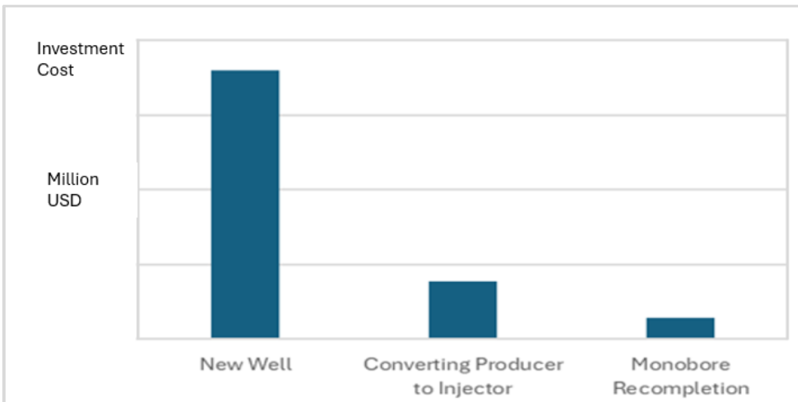

Figure 11—Cost Comparison of New Injector, Converting Producer to Injector and Monobore Recompletion in Merdeka Field

By using actual job cost and actual oil production response from the monobore pilot job in INJ-1, an economic review has been conducted. It indicates that the job is economic and feasible for further implementation in the field, as shown in Table 2.

Table 2—Economic Review of Monobore Recompletion Job in Merdeka Field

POT (year)	PI ($/$)
<1	2.56

An additional benefit of using monobore injector design is the potential significant decrease in future well intervention costs. The simplified well design allows many routine maintenance and operations to be performed using rigless intervention methods. By reducing the need for conventional workover rig, operators can cut both direct rig-related expenses and downtime, ultimately improving the efficiency of the waterflood operations.

Plan Forward

Given the positive pilot result, a forward plan has been developed to apply this method to other candidates with similar issues in Merdeka Field. Several candidates have been reviewed for the next phase of implementation with proper alignment with future field development plans. Following the successful pilot, the asset and drilling team from other mature waterflood field in the company have planned to implement monobore completion on the injectors in their respective fields.

Conclusions

Restoration of Injection and Enhanced Waterflood Performance

The injector chosen for the pilot study was selected based on a comprehensive evaluation of subsurface and surface data. Several key screening criteria were applied in selecting the pilot candidate, including low VRR performance, injector located at downdip or lower structural locations, strong reservoir connectivity, high producer to injector ratio and the absence of nearby wells with shallow fluid levels. Workover history and surface injection system conditions were also being carefully reviewed.

INJ-1, which was previously idle due to deep casing leak and considered as non-repairable through conventional methods, was successfully reactivated using a monobore remedial completion. Best practices from the pilot workover execution, such as the selection of wellhead with thread connection type, the utilization of 3-1/8" deep penetration gun with a density of 6 shots per foot for penetrating double casing, can be applied to future monobore injector implementations in Merdeka field.

Post-reactivation monitoring indicated a sustained injection rate, contributing to localized additional pressure support and improved injection and production performance in an area experiencing pressure decline. Improvements have also been observed in the surface production system, where several producer wells that were previously shut in due to surface facility constraints can be reactivated.

Improved Well Integrity and Cost Efficiency

The monobore well system, characterized by its simplified well design and single diameter casing profile, minimizes mechanical complexity and reduces the number of potential leaking points in conventional injectors e.g leakage from injection packers or leakage in casing annulus. The stable injection rate and pressure in the pilot well INJ-1 indicates improved wellbore integrity and enhanced long-term reliability of the injector, reducing the requirement for frequent well interventions in the future.

Compared to the cost of drilling new injectors or converting producers to injectors, the monobore recompletion offered a more economical alternative. The simplified well design also allows future routine maintenance to be performed using rigless intervention methods. Capital and operational expenditures can be minimized, enabling the operator to consider a broader injector reactivation program across the asset.

Challenges

While monobore injector designs simplify well completions, they can pose challenges for conducting specific zonal isolation in injectors with multiple injection intervals and highly heterogeneous reservoirs. A well-planned design, combined with strategic candidate and zonal selection are essential for effective implementation

The success of the INJ-1 reactivation demonstrates the potential field-wide application of monobore completions as part of a cost-effective reservoir pressure maintenance strategy. This case sets an alternative solution for other mature waterflood fields facing similar challenges. Strategic reuse of idle injectors with simplified completions and enhanced reliability can reduce the requirement for new injectors to support future field development plan

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